MIL-B-60075A(MU)
5 August 1966
SUPERSEDING
MIL-B-60075(MU)
18 June 1964

MILITARY SPECIFICATION

BOMB, NONPERSISTENT GAS, SPHERICAL, 4.5-INCH, M139; METAL PARTS

1. SCOPE

1.1 This specification covers inert spherical bomb components.

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

PPP-B-566	-	Boxes, Folding, Paperboard.
PPP-B-636	-	Box, Fiberboard.

MILITARY

MIL-P-116	-	Preservation, Methods of.
MIL-W-8604	-	Welding of Aluminum Alloys: Process For.

STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	-	Marking for Shipment and Storage.

FSC 1325

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DRAWINGS

US ARMY EDGEWOOD ARSENAL

DL14-5-2759 - Bomb, Nonpersistent Gas, Spherical, 4.5 Inch, M139, Metal Parts.
(All drawings referenced in this specification are listed on above drawing.)

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

UNIFORM CLASSIFICATION COMMITTEE

Uniform Freight Classification Ratings, Rules and Regulations.

(Application for copies of these freight classification rules should be addressed to the Uniform Classification Committee, 202 Union Station, Chicago, Illinois 60606.)

3. REQUIREMENTS

3.1 Materials and components.

3.1.1 Materials. All materials cited on Drawing DL14-5-2759 or on the subsidiary drawings shall conform to the specifications listed thereon, or to the specific characteristics set forth on the drawings.

3.1.2 Components. All components of the bomb shall conform to the specifications and drawings listed on Drawing DL14-5-2759 and subsidiary drawings.

3.2 Assembly. The shell halves shall be assembled as shown on Drawing C14-5-2760 and C14-5-2762.

3.3 Welding procedure. Prior to the start of any production, the supplier shall establish an approved welding procedure in accordance with MIL-W-8604, except that the paragraph describing maintenance sampling shall not be applicable. The welding procedures shall include provisions for repair and rework of weldment. The supplier shall submit with his proposed welding procedure the quality control procedures to be used during all the production to assure that the approved welding procedure is

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maintained and controlled, and is producing welds conforming to MIL-W-8604. The supplier shall requalify his procedure whenever it varies from the accepted welding procedure. Compliance with this requirement shall be in accordance with 4.3.4.3(a).

3.4 Rework. Only one weld repair shall be permitted per shell half (see 4.3.4.1).

3.5 Helium leakage. The shell half shall not leak helium at a rate greater than 1×10^{-6} atmospheric cubic centimeter (cc) per second when tested in accordance with 4.3.4.2.

3.6 Preproduction. Prior to the start of regular production, a preproduction sample of components shall be produced in accordance with this specification for examination and test (see 4.2).

3.7 Workmanship. The components shall be free from damage that could prevent their normal assembly and function.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.

4.1.1 Supplier's responsibility. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to specified requirements.

4.1.2 Objective evidence. The supplier shall provide objective evidence acceptable to the contracting officer that the requirements of 3.1 and section 5 for which specific inspection has not been provided in this specification have been satisfied.

4.2 Preproduction.

4.2.1 Lotting. Prior to the start of production, a preproduction lot of 200 bombs shall be produced using the same methods, materials, procedures, and equipment that will be used in regular production.

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4.2.2 Examination and test. Each bomb shall be examined and tested for all the requirements in this specification.

4.2.3 Acceptance/rejection criteria. The preproduction lot shall meet all the requirements listed in specification to be acceptable. The supplier shall obtain written approval from the contracting officer prior to proceeding with regular production.

4.3 Regular production.

4.3.1 Lotting. A lot shall consist of like components produced by one manufacturer, at one plant, and under essentially the same manufacturing conditions.

4.3.2 Sampling.

4.3.2.1 Tool controlled dimensions. At the beginning of regular production or following each "down time," one or more components shall be taken for examination from each die (not including the first or intermediate dies in a progressive series) that produces a tool controlled dimension (not a machined dimension) which appears in the end item. During regular production, the frequency and quantity of these examinations shall be at the discretion of the supplier, but shall not be less than one in every 8 hours of production.

4.3.2.2 Machined components. Sampling shall be conducted in accordance with MIL-STD-105.

4.3.3.3 For critical defects. Each item in the lot shall be inspected for its critical characteristics listed in the classification of defects.

4.3.3.4 Classification of defects.

(a) Shell halves (Dwg D14-5-2761 and D14-5-2763).

Categories	Defects
Critical:	None defined
Major:	AQL 1.0 percent defective
101	Shell wall thickness incorrect

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4.3.3.4 (a) Shell halves (Dwg D14-5-2761 and D14-5-2763). (Continued)

Categories	Defects
Minor:	AQL 2.5 percent defective
201	Flange diameter incorrect
202	Flange channel depth incorrect
203	Flange thickness incorrect
204	Hole diameter incorrect
205	Plug opening neck outside diameter incorrect
206	Fin clearance incorrect
207	Overall height incorrect
208	Workmanship (3.7)

(b) Plug (Dwg B14-5-3390).

Categories	Defects
Critical:	None defined
Major:	AQL 1.0 percent defective
101	Overall height incorrect
102	Datum diameter incorrect
103	Boss Height incorrect
104	Hole diameter incorrect
105	Boss outside diameter incorrect
106	Wall thickness incorrect
107	Finish (63) incorrect
Minor:	AQL 2.5 percent defective
201	Chamfer missing or incorrect

(c) Fastener (Dwg B14-5-2765).

Categories	Defects
Critical:	None defined
Major:	AQL 1.0 percent defective
101	Height inside (flange) incorrect
102	Cut out location incorrect
103	Radii 2 25/64 incorrect
104	Flange dimensions incorrect

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4.3.3.4 (c) Fastener (Dwg B14-5-2765). (Continued)

Categories	Defects
Minor:	AQL 2.5 percent defective
201	Burrs or sharp edges
202	Workmanship (3.7)
203	Protective finish missing or incorrect

(d) Shell halves (Dwgs C14-5-2760 and C14-5-2762).

Categories	Defects	Acceptance standards
Critical:		
1	Helium leakage	4.3.4.2
Major:	AQL 1.0 percent defective	
101	Plastic plug or cap missing	
102	Flatness of partition incorrect	
103	Fin alignment incorrect	

4.3.4 Tests. Tests shall be conducted as follows:

4.3.4.1 Rework. If the shell assembly (Dwgs C14-5-2760 and C14-5-2762) fails the helium leakage test, the assembly may be reworked by a procedure which shall be a part of the welding procedure requirements. Reworked assemblies shall be submitted in accordance with the requirements of this specification for acceptance in a separate lot and shall not be included in lots of assemblies which have been accepted.

4.3.4.2 Helium leakage. The shell half shall be tested for helium leakage using a Government approved helium leak testing device. If the shell is accepted, the letter "E" shall be stamped in ink near the filling port.

4.3.4.3 Welding.

(a) Procedure. Prior to performing any weldments on the bomb, the contractor shall have submitted and received approval subject to "Qualification"

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of the welding procedure which he proposes to use in accordance with the requirements of MIL-W-8604. In additon the contractor shall include the cleaning and rework procedure which he proposes to use to produce weldments on the bomb meeting the requirements of MIL-W-8604 and contract requirements. Approval of the welding rework and cleaning procedure shall be made by the contracting officer of the procuring activity.

(b) Qualification. The supplier shall produce a qualification lot of 50 bomblet half assemblies or more as specified in each contract using the proposed welding procedures to demonstrate the satisfactory quality of the weld produced. Evaluation and approval of the proposed procedures shall be based upon the following:

(1) Review of written procedures (including cleaning and timing of sequences).

(2) Microscopic and macroscopic examination for:

(a) Fusing and joint penetration
(b) Cracks
(c) Porosity and inclusions
(d) Overlapping
(e) Other metalic discontingencies

(3) Helium testing of closure

5. PREPARATION FOR DELIVERY

5.1 Packaging. Levels shall be as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Plug. Fifty plugs shall be placed in a paperboard folding box conforming to PPP-B-566 and packaged in accordance with method IC-1 or IC-3 of MIL-P-116.

5.1.1.2 Fastener. Fifty fasteners shall be packaged in a W6c fiberboard box conforming to PPP-B-636. Fasteners shall be restricted from free movement. Box shall be closed and all seams and joints sealed with tape in accordance with appendix to PPP-B-636.

5.1.1.3 Right shell half and left shell half assemblies. The opening in each assembly shall be covered with either a plastic cap or plug.

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5.1.2 Level C. Like components of the bomb shall be packaged together to afford adequate protection from the supply source to the first receiving activity for immediate use.

5.2 Packing. Levels shall be as specified (see 6.2).

5.2.1 Level A.

5.2.1.1 Plug. Plugs packaged as specified in 5.1.1.1 shall be packed not to exceed the weight limitations in a V3c, style optional fiberboard container conforming to PPP-B-636. In addition to the specified closure requirements, box seams and manufacturer's joint shall be taped. When metal banding is used, it shall be galvanized.

5.2.1.2 Fasteners. Fasteners packaged as specified in 5.1.1.2 shall be packed in the same manner as specified in 5.2.1.1.

5.2.1.3 Right shell half and left shell half assemblies. Shell half assemblies of one kind packaged as specified in 5.1.1.3 shall be packed together in the manner as specified in 5.2.1.1, except that the assemblies shall be packed in layers, one per compartment (snug fitting), formed by interlocking slotted fiberboard partitions and pads shall be fabricated from the same material as the box.

5.2.2 Level C. Like components of the bomb packaged as specified in 5.1.2 shall be packed together in a manner to insure carrier acceptance and safe delivery at first domestic destination. Containers shall be in accordance with Uniform Freight Classification rules or the regulations of other carriers applicable to the mode of transportation.

5.3 Marking. In addition to any special marking required by the contract or order, unit packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. These components are used in the manufacture of the M139 bomb.

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6.2 Ordering data. Procurement documents should specify the following:

(a) The title, number and date of this specification.
(b) Level of packaging and packing required.
(c) Preproduction.

(1) Time allowed for suppliers submission of samples for Government test and evaluation after award of contract.

(2) Name and address of test facility and shipping instructions when testing is performed by the Government.

(3) Time required for the Government to notify the supplier whether or not to proceed with production.

Custodian:

Army - MU

Preparing activity:

Army - MU(EA)

Project No 1325-A167

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)* | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4663, WHICH IS OBSOLETE PLATE NO. 15419

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